

# **PROBABILITY III**

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## Chapter 1

# REFINING THE NOTION OF PROBABILITY

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Consider the random walk on the line; a drunk ant sits at the origin and takes steps of length 1 to the left or right with equal probability. We wish to obtain the probability that after an even  $n$  steps, the ant returns to the origin. Of course, the natural approach is to frame this problem in a discrete probability space.

**Definition 1.1.** A *discrete probability space* is a pair  $(\Omega, q)$  where  $\Omega$  is a non-empty finite or countably infinite set called the sample space, and  $q : \Omega \rightarrow [0, 1]$  is a function satisfying  $\sum_{\omega \in \Omega} q(\omega) = 1$ . Any subset of  $\Omega$  is termed an event, and the probability of an event  $A \subseteq \Omega$  is defined as  $\mathbb{P}(A) = \sum_{\omega \in A} q(\omega)$ .

For the problem above, we can take the sample space to be

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \dots, x_n) \in \mathbb{Z}^{n+1} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } 0 \leq i < n\}, \quad (1.1)$$

and the mapping  $q$  to be  $q(\mathbf{x}) = 2^{-n}$  for all  $\mathbf{x} \in \Omega$ . Then, the event that the ant returns to the origin after  $n$  steps is given by  $A = \{\mathbf{x} \in \Omega \mid x_n = 0\}$ , and we have

$$\mathbb{P}(A) = \sum_{\mathbf{x} \in A} q(\mathbf{x}) = 2^{-n} |A| = 2^{-n} \binom{n}{n/2}. \quad (1.2)$$

Reframing the problem, let us consider a random walk on the line starting at the origin, and continuing indefinitely. In this case, the sample space is given by

$$\Omega = \{\mathbf{x} = (x_0 = 0, x_1, x_2, \dots) \in \mathbb{Z}^{\mathbb{N}} \mid x_{i+1} - x_i \in \{-1, 1\} \text{ for all } i \in \mathbb{N}\}. \quad (1.3)$$

This  $\Omega$  is uncountably infinite. In such a case, we cannot define a probability measure by assigning a positive probability to each element of  $\Omega$ . As another example of difficulty in dealing with uncountable sample spaces, consider the problem of choosing a real number uniformly at random from the interval  $[0, 1]$ . If we attempt to assign a positive probability to each element of  $[0, 1]$ , we would have to assign a probability of 0 to each element, since there are uncountably many elements in  $[0, 1]$ . However, this would lead to a total probability of 0, which is not valid. Of course, we know that the probability of choosing a specific element or a countable set of elements from  $[0, 1]$  is 0, but there then arises the question of how to assign a probability to an uncountable set of elements, such as the Cantor set. We can work in a slightly different viewpoint.

**Definition 1.2.** A *probability space* is a tuple  $(\Omega, \mathcal{F}, \mathbb{P})$  where  $\Omega$  is a non-empty set called the sample space,  $\mathcal{F}$  is the collection of all subsets of  $\Omega$  ( $\sim 2^{\Omega}$ ), and  $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$  is a function satisfying  $\mathbb{P}(\Omega) = 1$  and countable additivity, that is, for any countable collection of pairwise disjoint sets  $A_1, A_2, \dots \in \mathcal{F}$ ,

we have

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i). \quad (1.4)$$

Using this definition, we look to define a  $\mathbb{P}$  on subsets of  $[0, 1]$  that satisfies the above properties, along with the property that  $\mathbb{P}([a, b]) = b - a$ . Thus, we have  $\mathbb{P}([0, 1]) = 1$ . If we have two disjoint intervals  $I_1, I_2 \subseteq [0, 1]$ , then  $\mathbb{P}(I_1 \cup I_2) = \mathbb{P}(I_1) + \mathbb{P}(I_2)$ . This is a natural property to expect from a probability measure. What about a general set  $S \subseteq [0, 1]$ ? One can observe that if we have a sequence of intervals  $I_1, I_2, \dots$  such that  $S \subseteq \bigcup_{k=1}^{\infty} I_k$ , then we would have  $\mathbb{P}(S) \leq \sum_{k=1}^{\infty} \mathbb{P}(I_k)$ . This is because the probability of  $S$  should be less than or equal to the probability of any set that contains  $S$ . Thus, we can define

$$\mathbb{P}_*(S) = \inf \left\{ \sum_{k=1}^{\infty} |I_k| \mid S \subseteq \bigcup_{k=1}^{\infty} I_k, I_k \subseteq [0, 1] \right\}. \quad (1.5)$$

We have not verified that this is a valid definition of a probability measure, hence the subscript  $*$ . Via this, let us try computing  $\mathbb{P}_*(\mathbb{Q} \cap [0, 1])$ . Enumerate the rationals in  $[0, 1]$  as  $\{q_1, q_2, \dots\}$ . For any  $\epsilon > 0$ , we can choose intervals  $I_k = [q_k - \epsilon/2^{k+1}, q_k + \epsilon/2^{k+1}] \cap [0, 1]$  for each  $k \in \mathbb{N}$ . Note that  $\bigcup_{k=1}^{\infty} I_k$  contains all the rationals in  $[0, 1]$ , and  $\sum_{k=1}^{\infty} |I_k| \leq \sum_{k=1}^{\infty} \epsilon/2^k = \epsilon$ . Since  $\epsilon$  was arbitrary, we have  $\mathbb{P}_*(\mathbb{Q} \cap [0, 1]) = 0$ . This is consistent with our intuition that the probability of choosing a rational number from  $[0, 1]$  is 0. So,  $\mathbb{P}_*$  seems to be a reasonable definition of a probability measure on  $([0, 1], 2^{[0, 1]})$ . However, we will show later that there exists a set  $\mathcal{A} \subseteq [0, 1]$  such that  $\mathbb{P}_*(\mathcal{A}) = \mathbb{P}_*([0, 1] \setminus \mathcal{A}) = 1$ , contradicting the property that  $\mathbb{P}_*([0, 1]) = 1$ . We state the following theorem without proof.

**Theorem 1.3.** *There is no probability measure  $\mathbb{P}$  on  $([0, 1], 2^{[0, 1]})$  such that  $\mathbb{P}([a, b]) = b - a$  for all  $[a, b] \subseteq [0, 1]$ .*

If we remove the requirement that the probability measure of a subinterval  $[a, b]$  is equal to its length, then we can define a probability measure on  $([0, 1], 2^{[0, 1]})$ . For example, we can define  $\mathbb{P}(S) = 1$  if  $1/2 \in S$  and  $\mathbb{P}(S) = 0$  otherwise. This is a valid probability measure, but it is not very useful. Keeping the requirement that the probability measure of a subinterval  $[a, b]$  is equal to its length, we can instead restrict the collection of subsets of  $[0, 1]$  on which we define the probability measure.

**Definition 1.4.** A *probability space* is a tuple  $(\Omega, \mathcal{F}, \mathbb{P})$  where

- $\Omega$  is a non-empty set called the sample space,
- $\mathcal{F}$  is a collection of subsets of  $\Omega$  satisfying the following properties:
  - $\Omega \in \mathcal{F}$ ,
  - if  $A \in \mathcal{F}$ , then  $\Omega \setminus A \in \mathcal{F}$ ,
  - if  $A_1, A_2, \dots \in \mathcal{F}$ , then  $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ ,
- $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$  is a function satisfying:
  - $\mathbb{P}(\Omega) = 1$ ,
  - if  $A_1, A_2, \dots \in \mathcal{F}$  are pairwise disjoint, then  $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$ .

Such an  $\mathcal{F}$  is termed a  $\sigma$ -algebra on  $\Omega$ , and such a  $\mathbb{P}$  is termed a *probability measure* on  $(\Omega, \mathcal{F})$ .

## 1.1 $\sigma$ -algebras

**Example 1.5.** Given  $\Omega$ , trivial possible choices for  $\mathcal{F}$  are  $\{\emptyset, \Omega\}$  and  $2^\Omega$ . A more interesting example is the following: suppose  $\Omega$  is uncountable. Then  $\mathcal{F} = \{A \subseteq \Omega \mid A \text{ is countable or } \Omega \setminus A \text{ is countable}\}$  is a  $\sigma$ -algebra on  $\Omega$ .

Let  $S \subseteq 2^\Omega$  be a collection of subsets of  $\Omega$ . We claim that there exists a notion of the smallest  $\sigma$ -algebra containing  $S$ . This is called the  $\sigma$ -algebra generated by  $S$ , and is denoted by  $\sigma(S)$ .

**Lemma 1.6.** *For any collection  $S \subseteq 2^\Omega$ , there exists a unique smallest  $\sigma$ -algebra containing  $S$  satisfying*

$$\sigma(S) = \bigcap \{ \mathcal{F} \subseteq 2^\Omega \mid \mathcal{F} \text{ is a } \sigma\text{-algebra and } S \subseteq \mathcal{F} \}. \quad (1.6)$$

*Proof.* It is easy to check that  $\sigma(S)$  is indeed a  $\sigma$ -algebra containing  $S$ . If  $\mathcal{F}$  is any  $\sigma$ -algebra containing  $S$ , then  $\sigma(S) \subseteq \mathcal{F}$ . Thus,  $\sigma(S)$  is the unique smallest  $\sigma$ -algebra containing  $S$ . ■

**Definition 1.7.** Let  $(X, d)$  be a metric space (or a topological space), and let  $\mathcal{S} \subseteq 2^X$  be the collection of all open sets in  $X$ . Then, the  $\sigma$ -algebra  $\sigma(\mathcal{S})$  is called the *Borel  $\sigma$ -algebra* on  $X$ , and its elements are called *Borel sets*.

We will show that there exists a probability measure on  $([0, 1], \mathcal{B})$ , where  $\mathcal{B}$  is the Borel  $\sigma$ -algebra on  $[0, 1]$ , which extends the notion of length of intervals to Borel sets.

Let us return to the problem of indefinite random walks on the line. Let  $\Omega = \{\omega = (\omega_1, \omega_2, \dots) \mid \omega_i \in \{0, 1\}\}$ , where  $\omega_i = 0$  corresponds to a step to the left and  $\omega_i = 1$  corresponds to a step to the right. If we let  $A = \{0, 1\}$ , then  $\Omega = A^\mathbb{N}$ ; consider the discrete topology on  $A$ , and the product topology on  $\Omega$ . If we let

$$\mathcal{S} = \{U_1 \times U_2 \times \dots \times U_k \times A \times A \times \dots \mid U_i = \{0\} \text{ or } \{1\}\} \quad (1.7)$$

then one can show that the  $\sigma$ -algebra generated by  $\mathcal{S}$  is the Borel  $\sigma$ -algebra on  $\Omega$ . We can set  $\mathcal{F} = \sigma(\mathcal{S})$ , and define a probability measure  $\mathbb{P}$  on  $(\Omega, \mathcal{F})$  by setting  $\mathbb{P}(U_1 \times U_2 \times \dots \times U_k \times A \times A \times \dots) = 2^{-k}$  for all  $U_i = \{0\}$  or  $\{1\}$ .

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**Example 1.8.** Let  $(X, d)$  be a separable metric space (that is, there exists a countable dense subset). The Borel  $\sigma$ -algebra on  $X$  is then

$$\mathcal{B}(X) = \sigma(\{B(p, r)\}_{p \in X, r > 0}). \quad (1.8)$$

Note that the usual containment is  $\sigma(\{B(p, r)\}_{p \in X, r > 0}) \subseteq \mathcal{B}(X)$  for a metric space  $X$ . So, it suffices to show that if  $V \in \tau$ , the topology on  $X$ , then  $V \in \sigma(\{B(p, r)\}_{p \in X, r > 0})$ . Let  $Q$  be a countable dense subset of  $X$ . Then, for any  $V \in \tau$ , we have

$$V = \bigcup_{n=1}^{\infty} B(q_n, r_n), \quad (1.9)$$

where  $q_n \in Q \cap V$  and  $r_n > 0$  (where  $r_n$  is the largest possible radius such that  $B(q_n, r_n) \subseteq V$ ). Since  $\sigma(\{B(p, r)\}_{p \in X, r > 0})$  is a  $\sigma$ -algebra, we have  $V \in \sigma(\{B(p, r)\}_{p \in X, r > 0})$ . Thus, we have  $\mathcal{B}(X) = \sigma(\{B(p, r)\}_{p \in X, r > 0})$ .

**Example 1.9.** Let  $(\Sigma, \mathcal{F}, \mathbb{P})$  be a probability space.

- $\mathcal{F}$  is closed under countable unions, countable intersections, differences, and symmetric differences. Countable unions and intersections follow from the definition of a  $\sigma$ -algebra. If  $A, B \in \mathcal{F}$ , then  $A \setminus B = A \cap (\Omega \setminus B) \in \mathcal{F}$ . Finally,  $A \triangle B = (A \setminus B) \cup (B \setminus A) \in \mathcal{F}$ .
- Let  $(\Sigma, \mathcal{F}, \mathbb{P})$  be a probability space, and let  $A_1, A_2, \dots \in \mathcal{F}$ . Define

$$\limsup_{n \rightarrow \infty} A_n := \{x \in \Omega \mid x \text{ belongs to infinitely many } A_n\} \quad (1.10)$$

and

$$\liminf_{n \rightarrow \infty} A_n := \{x \in \Omega \mid x \text{ belongs to all but finitely many } A_n\}. \quad (1.11)$$

Then, both the  $\limsup$  and  $\liminf$  belong in  $\mathcal{F}$ . To see this, simply note that

$$\limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k, \quad \liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k. \quad (1.12)$$

- One may show that  $\mathbb{P}(\emptyset) = 0$ ,  $\mathbb{P}(A) \leq \mathbb{P}(B)$  if  $A \subseteq B$ ,  $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ , and  $\mathbb{P}(\bigcup A_n) \leq \sum \mathbb{P}(A_n)$  for any countable collection of sets  $A_1, A_2, \dots \in \mathcal{F}$ .
- For a nested sequence of sets  $A_1 \subseteq A_2 \subseteq \dots$  in  $\mathcal{F}$ , let  $A = \bigcup_{n=1}^{\infty} A_n$  (this is usually denoted by  $A_n \uparrow A$ ). Then,  $\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$ . To this end, let  $B_n = A_n \setminus A_{n-1}$  for  $n > 1$ , and  $B_1 = A_1$ . Then  $A = \bigcup_{n=1}^{\infty} B_n$ . Thus,

$$\mathbb{P}(A) = \sum_{n=1}^{\infty} \mathbb{P}(B_n) = \lim_{N \rightarrow \infty} \sum_{n=1}^N \mathbb{P}(B_n) = \lim_{N \rightarrow \infty} \sum_{n=2}^N \mathbb{P}(A_n) - \mathbb{P}(A_{n-1}) + \mathbb{P}(A_1) = \lim_{N \rightarrow \infty} \mathbb{P}(A_N). \quad (1.13)$$

### 1.1.1 Other Systems of Sets

The following are other interesting collection of subsets of  $\Omega$  besides the Borel  $\sigma$ -algebra.

1.  $\pi$ -system: a collection of subsets  $S \subseteq \Omega$  such that  $A, B \in S$  implies  $A \cap B \in S$ .
2.  $\lambda$ -system: a collection  $\Lambda$  of subsets of  $\Omega$  such that  $\Omega \in \Lambda$ , if  $A, B \in \Lambda$  with  $A \subseteq B$ , then  $B \setminus A \in \Lambda$ , and if  $A_1, A_2, \dots \in \Lambda$ , then  $\bigcup_{n=1}^{\infty} A_n \in \Lambda$ .
3. algebra: a collection  $\mathcal{A}$  of subsets of  $\Omega$  such that  $\Omega \in \mathcal{A}$ , if  $A \in \mathcal{A}$ , then  $\Omega \setminus A \in \mathcal{A}$ , and if  $A, B \in \mathcal{A}$ , then  $A \cup B \in \mathcal{A}$ .

We had defined  $\omega(S)$ , the  $\sigma$ -algebra generated by a collection of subsets  $S \subseteq 2^\Omega$ . We can similarly define  $\pi(S)$ , the  $\pi$ -system generated by  $S$ ,  $\lambda(S)$ , the  $\lambda$ -system generated by  $S$ , and  $\mathcal{A}(S)$ , the algebra generated by  $S$ .

**Theorem 1.10** (*Sierpiński-Dynkin  $\pi$ - $\lambda$  Theorem*). *Let  $S, \mathcal{F} \subseteq 2^\Omega$  be collections of subsets of  $\Omega$ . Then,*

1.  $\mathcal{F}$  is a  $\sigma$ -algebra if and only if  $\mathcal{F}$  is both a  $\pi$ -system and a  $\lambda$ -system.
2. If  $S$  is a  $\pi$ -system, then  $\lambda(S) = \sigma(S)$ .

*Proof.* 1. If  $\mathcal{F}$  is a  $\sigma$ -algebra, then it is trivially both a  $\pi$ -system and a  $\lambda$ -system. Conversely, if  $\mathcal{F}$  is both a  $\pi$ -system and a  $\lambda$ -system, we need to verify the axioms of a  $\sigma$ -algebra. We have  $\Omega \in \mathcal{F}$  since  $\mathcal{F}$  is a  $\lambda$ -system. If  $A \in \mathcal{F}$ , then  $\Omega \setminus A \in \mathcal{F}$  since  $\mathcal{F}$  is a  $\lambda$ -system. Finally, if  $A_1, A_2, \dots \in \mathcal{F}$ , then we can write  $B_n = \bigcup_{i=1}^n A_i \in \mathcal{F}$  since  $A_i^c \in \mathcal{F}$  and  $B_n^c = \bigcap_{i=1}^n A_i^c \in \mathcal{F}$  since  $\mathcal{F}$  is a  $\pi$ -system. Then,  $B_n \uparrow B = \bigcup_{i=1}^{\infty} A_i$ , and since  $\mathcal{F}$  is a  $\lambda$ -system, we have  $B \in \mathcal{F}$ . Thus,  $\mathcal{F}$  is a  $\sigma$ -algebra.

2. Let  $S$  be a  $\pi$ -system. Since  $\lambda(S) \subseteq \sigma(S)$ , it suffices to show that  $\lambda(S)$  is a  $\pi$ -system. Fix  $A \in S$ , and define  $\mathcal{C}_A = \{B \in \lambda(S) \mid A \cap B \in \lambda(S)\}$ . Note that  $S \subseteq \mathcal{C}_A$  because  $S$  is a  $\pi$ -system. We show that  $\mathcal{C}_A$  is a  $\lambda$ -system. We have  $\Omega \in \mathcal{C}_A$  since  $A \cap \Omega = A \in \lambda(S)$ . If  $B_1, B_2 \in \mathcal{C}_A$  with  $B_1 \subseteq B_2$ , then  $A \cap B_1 \subseteq A \cap B_2$ , and since  $A \cap B_2 \in \lambda(S)$ , we have  $(A \cap B_2) \setminus (A \cap B_1) = A \cap (B_2 \setminus B_1) \in \lambda(S)$ , and thus  $B_2 \setminus B_1 \in \mathcal{C}_A$ . Finally, if  $B_1, B_2, \dots$  are in  $\mathcal{C}_A$ , then  $A \cap B_n \in \lambda(S)$  for all  $n$ , and thus  $\bigcup (A \cap B_n) = A \cap (\bigcup B_n) \in \lambda(S)$ , and hence  $\bigcup B_n \in \mathcal{C}_A$ . Thus,  $\mathcal{C}_A$  is a  $\lambda$ -system. Since  $\lambda(S)$  is the smallest  $\lambda$ -system containing  $S$ , we have  $\lambda(S) = \mathcal{C}_A$ , and hence for any  $B, C \in S$ , we have  $B, C \in S$  implies  $B \cap C = A \cap C \in S$ . Thus,  $\lambda(S)$  is a  $\pi$ -system. ■

**Lemma 1.11.** *Let  $S$  be a  $\pi$ -system, and let  $\mathbb{P}$  and  $\mathbb{Q}$  be two probability measures on  $(\Sigma, \sigma(S))$  such that  $\mathbb{P}(A) = \mathbb{Q}(A)$  for all  $A \in S$ . Then,  $\mathbb{P}(A) = \mathbb{Q}(A)$  for all  $A \in \sigma(S)$ .*

*Proof.* Let  $\mathcal{C} = \{B \in \sigma(S) \mid \mathbb{P}(B) = \mathbb{Q}(B)\}$ . Then,  $S \subseteq \mathcal{C}$ , and we show that  $\mathcal{C}$  is a  $\lambda$ -system. We have  $\Omega \in \mathcal{C}$  since  $\mathbb{P}(\Omega) = 1 = \mathbb{Q}(\Omega)$ . If  $B_1, B_2 \in \mathcal{C}$  with  $B_1 \subseteq B_2$ , then  $\mathbb{P}(B_2 \setminus B_1) = \mathbb{P}(B_2) - \mathbb{P}(B_1) = \mathbb{Q}(B_2) - \mathbb{Q}(B_1) = \mathbb{Q}(B_2 \setminus B_1)$ , and hence  $B_2 \setminus B_1 \in \mathcal{C}$ . Finally, if  $B_1, B_2, \dots$  are in  $\mathcal{C}$ , then  $\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$ , and since  $\mathbb{Q}(A) = \lim_{n \rightarrow \infty} \mathbb{Q}(A_n)$ , we have  $\mathbb{P}(A) = \mathbb{Q}(A)$ . ■

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**Example 1.12.** We wish to obtain an example of  $S$  which is *not* a  $\pi$ -system, and two probability measures  $\mathbb{P}$  and  $\mathbb{Q}$  on  $(\Sigma, \sigma(S))$  such that  $\mathbb{P}(A) = \mathbb{Q}(A)$  for all  $A \in S$ , but  $\mathbb{P}(A) \neq \mathbb{Q}(A)$  for some  $A \in \sigma(S)$ . Let  $\Omega = \{1, 2, 3, 4\}$ , and let  $S = \{\{1, 2\}, \{2, 3\}\}$ . Then,  $\sigma(S) = 2^\Omega$ . Define  $\mathbb{P}$  and  $\mathbb{Q}$  on  $(\Omega, \sigma(S))$  by

$$\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \mathbb{P}(\{4\}) = 1/4, \quad (1.14)$$

and

$$\mathbb{Q}(\{1\}) = \mathbb{Q}(\{3\}) = 1/6, \quad \mathbb{Q}(\{2\}) = \mathbb{Q}(\{4\}) = 1/3. \quad (1.15)$$

Then,  $\mathbb{P}(\{1, 2\}) = \mathbb{Q}(\{1, 2\}) = 1/2$  and  $\mathbb{P}(\{2, 3\}) = \mathbb{Q}(\{2, 3\}) = 1/2$ , but  $\mathbb{P}(\{3\}) = 1/4 \neq 1/6 = \mathbb{Q}(\{3\})$ .

**Example 1.13.** Let  $\Omega \neq \emptyset$ , and  $A_\lambda$  be a collection of subsets of  $\Omega$  indexed by  $\lambda \in \Lambda$ . Consider the  $\sigma$ -algebra  $\sigma(\{A_\lambda\}_{\lambda \in \Lambda})$ , and pick  $B \in \sigma(\{A_\lambda\}_{\lambda \in \Lambda})$ . We will show that there exists a countable subcollection  $\{A_{\lambda_n}\}_{n=1}^\infty$  such that  $B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty)$ . Let  $\mathcal{C} = \{B \in \sigma(\{A_\lambda\}_{\lambda \in \Lambda}) \mid B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty) \text{ for some countable subcollection } \{A_{\lambda_n}\}_{n=1}^\infty\}$ . Then,  $\mathcal{C}$  is a  $\sigma$ -algebra containing  $\{A_\lambda\}_{\lambda \in \Lambda}$ , and hence  $\sigma(\{A_\lambda\}_{\lambda \in \Lambda}) \subseteq \mathcal{C}$ . Thus,  $B \in \mathcal{C}$ , and hence there exists a countable subcollection  $\{A_{\lambda_n}\}_{n=1}^\infty$  such that  $B \in \sigma(\{A_{\lambda_n}\}_{n=1}^\infty)$ .

## 1.2 The Lebesgue Measure

**Theorem 1.14.** *There exists a unique Borel measure  $\lambda$  on  $[0, 1]$  such that  $\lambda(I) = |I|$  for all intervals  $I \subseteq [0, 1]$ .*

*Proof.* For the  $\pi$ -system  $S = \{(a, b) \mid 0 \leq a < b \leq 1\}$ , we know that  $\sigma(S)$  is a Borel  $\sigma$ -algebra on  $[0, 1]$ . If  $\lambda$  and  $\mu$  are two Borel measures on  $[0, 1]$  such that  $\lambda(I) = \mu(I) = |I|$  for all intervals  $I \subseteq [0, 1]$ , then by the previous lemma, we have  $\lambda(A) = \mu(A)$  for all  $A \in \sigma(S)$ . Thus, the Borel measure is unique.

To show the existence of such a measure, we will define a pseudomeasure  $\lambda_*$  that satisfies the interval property, and then show that it is a measure on the Borel  $\sigma$ -algebra. For any  $E \subseteq [0, 1]$ , define

$$\lambda_*(E) = \inf \left\{ \sum_{i=1}^\infty |I_i| \mid I_i \text{ are intervals and } E \subseteq \bigcup_{i=1}^\infty I_i \right\} \quad (1.16)$$

Infer that  $0 \leq \lambda_*(E) \leq 1$  for all  $E \subseteq [0, 1]$ . Moreover, let  $A$  and  $B$  be two sets in  $[0, 1]$ . Then, for any  $\varepsilon > 0$ , we have

$$\lambda_*(A \cup B) \leq \sum_{k=1}^\infty |I_k| + \sum_{k=1}^\infty |J_k| \leq \lambda_*(A) + \varepsilon + \lambda_*(B) + \varepsilon \implies \lambda_*(A \cup B) \leq \lambda_*(A) + \lambda_*(B). \quad (1.17)$$

Finally, let  $(a, b) \subseteq [0, 1]$  be an interval. Then,  $\lambda_*((a, b)) = b - a$ . To see this, suppose  $(a, b) \subseteq \bigcup_{k=1}^\infty I_k$  for some intervals  $I_k$ . Now consider  $[a + \varepsilon, b]$  for some  $\varepsilon > 0$  such that  $[a + \varepsilon, b] \subseteq \bigcup_{k=1}^\infty I_k$ . This is a compact subinterval, so we can extract a finite subcover  $[a + \varepsilon, b] \subseteq \bigcup_{k=1}^N I_k$ . Then,

$$b - a - \varepsilon = |[a + \varepsilon, b]| \leq \sum_{k=1}^N |I_k| \leq \sum_{k=1}^\infty |I_k| \implies b - a \leq \lambda_*((a, b)). \quad (1.18)$$

Now, since  $(a, b] \subseteq (a, b + \varepsilon)$ , a similar line of reasoning shows that  $\lambda_*((a, b]) \leq b - a$ . Thus, we have  $\lambda_*((a, b]) = b - a$ . The property of finite subadditivity also implies the property of countable subadditivity (show this). We now construct the  $\sigma$ -algebra  $\mathcal{F}$  on which  $\lambda_*$  is a measure. We pick

$$\mathcal{F} = \{A \subseteq [0, 1] \mid \lambda_*(E) = \lambda_*(E \cap A) + \lambda_*(E \cap A^c) \text{ for all } E \subseteq [0, 1]\}. \quad (1.19)$$

We show that  $\mathcal{F}$  is a  $\sigma$ -algebra containing the Borel  $\sigma$ -algebra, and that  $\lambda_*$  is a measure on  $\mathcal{F}$ . It is left as a minor exercise to show that  $\mathcal{F}$  is closed under intersections and complements (making  $\mathcal{F}$  a  $\pi$ -system). We show  $\mathcal{F}$  is a  $\lambda$ -system. Let  $A, B \in \mathcal{F}$ . Then  $B \setminus A = B \cap A^c$  is in  $\mathcal{F}$ . Then, for any  $E \subseteq [0, 1]$ , we have

$$\lambda_*(E) = \lambda_*(E \cap A_n) + \lambda_*(E \cap A_n^c) \quad (1.20)$$

$$\geq \lambda_*(E \cap A_n) + \lambda_*(E \cap A^c), \quad (1.21)$$

because  $A^c \subseteq A_n^c$ . Since  $A_n \uparrow A$ , we have  $E \cap A_n \uparrow E \cap A$ , and hence

$$\lambda_*(E \cap A_n) \uparrow \lambda_*(E \cap A).$$

Passing to the limit gives

$$\lambda_*(E) \geq \lambda_*(E \cap A) + \lambda_*(E \cap A^c).$$

On the other hand, subadditivity of the outer measure gives

$$\lambda_*(E) \leq \lambda_*(E \cap A) + \lambda_*(E \cap A^c).$$

Therefore,

$$\lambda_*(E) = \lambda_*(E \cap A) + \lambda_*(E \cap A^c),$$

so  $A \in \mathcal{F}$ . ■



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